

# Project Debriefing: Decentralized Dataspace Implementation

*By team Pink Panther*

**Project Team:** Dataspaces Project Team within Open Learning for ICT Fontys UAS

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## Version Control

| Version | Date  | Author       | Amendments       | Status                               |
|---------|-------|--------------|------------------|--------------------------------------|
| 1.0     | 05.03 | Pink Panther | Initial creation | First Draft for Stakeholder Feedback |

# 1. Introduction

## 1.1 Stakeholders

### Supervisory & Higher-Level Stakeholders

- **Mark M.J. Klerkx:** Stakeholder that we work directly under.
- **Toon A.W.J. Teeuwen:** Stakeholder that we work directly under.
- **Bart L.W.C. van Gennip:** Stakeholder positioned above Mark and Toon, though not directly in charge of the project team.

### Associated Entities & End Users

- **Data Space Support Centre (DSSC):** An overarching entity referenced in the project's foundational reading guide.
- **Customers and Companies:** Entities identified in the project requirements that need functional connectors (EDC/FIWARE) to connect and communicate with the dataspace.
- **Fontys:** The institution associated with the project ("Fontys FOR SOCIETY") and the provider of the VPN and Netlab infrastructure used for local deployment.

## 1.2 Cause and Problem Description

The project is driven by the transition from centralized data silos to decentralized "Dataspaces." The core problem is that organizations currently lack a standardized, secure way to share data while maintaining absolute sovereignty. Simply "connecting" is not enough; the connection must be governed by verifiable identity and strict usage policies to prevent unauthorized data exploitation.

## 2. Client Analysis

### 2.1 Client's Mission

The mission is to foster a collaborative ecosystem where data can be exchanged freely between trusted partners without the need for a central intermediary. The goal is to build a bespoke dataspace that demonstrates the practical application of data sovereignty.

### 2.2 Wants, Needs, and Wishes

- **Need:** Functional connectors for both the Customer and the Company to join the dataspace.
- **Want:** A comparison between two major framework architectures: **Eclipse Dataspace Components (EDC)** and **FIWARE**.
- **Wish:** To move beyond "dummy" services and implement real-world containers that mirror production environments.

### 2.3 Tangible Outcomes and Products

1. **Technical Deployment:** Functional EDC and FIWARE environments deployed locally on Netlab.
2. **Implementation Guide:** Comprehensive technical documentation (e.g., the FIWARE local deployment guide).
3. **Use Case Validation:** A successful demonstration of a data transfer governed by verifiable credentials.
4. **Role Division:** A clear group organizational structure finalized after technical testing.

## 3. Context Analysis

### 3.1 Challenge Context

The project operates in the emerging field of European Data Spaces (e.g., Gaia-X, IDS). The primary challenge is balancing technical connectivity with legal and ethical data governance.

### 3.2 Market Analysis

Currently, two major open-source philosophies dominate the market:

- **Eclipse Foundation (EDC):** Focuses on a "code-first," highly extensible Java framework.
- **FIWARE:** Focuses on a "container-first," modular ecosystem using standardized components.

The market is shifting toward "Trust Anchors"—services that allow companies to trust each others' employees based on cryptographically signed attributes.

### 3.3 Target Audience

The primary users are **system architects and data officers** at participating organizations who need to manage data access rights for their employees.

## 4. IT Analysis

### 4.1 Trends and Developments

The most significant trend in IT identity management is the move toward **W3C Standards**:

- **DID (Decentralized Identifiers)**: Acting as a digital "Passport."
- **VC (Verifiable Credentials)**: Acting as a digital "Driver's License."
- **ODRL (Open Digital Rights Language)**: Machine-readable contracts for usage control.

### 4.2 Existing Solutions

Existing solutions often rely on proprietary VPNs or centralized cloud storage. This project specifically analyzes **Eclipse EDC** (Java-heavy, modular) and **FIWARE Data Space Connector** (Docker/Container-based) as the state-of-the-art open-source alternatives.

## 5. Conclusion & Design Challenge

Based on the analysis, the project will move from a general research phase into a targeted implementation of a trust-based infrastructure.

### Design Challenge:

"Design a **Decentralized Dataspace Infrastructure** to enable **Organizations and their Employees** in a **Collaborative Data Ecosystem** to **Securely Exchange and Govern Data** in a **Policy-Driven, Sovereignty-Preserving Environment**."

## 6. Project Plan (v001) Preview

Following the approval of this debriefing, the project will proceed into three main phases:

1. **Technical Phase:** Setting up infrastructure (EDC & FIWARE) on Fontys VPN.
2. **Functional Phase:** Executing high-priority use cases using ODRL and VC.
3. **Finalization Phase:** Replacing dummy services with production-ready containers.

The team will synchronize every Monday at 09:00 on campus and Thursday afternoons to ensure progress.